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EXPLAIN COMPUTER ARCHITECTURE CONCEPTS

US 14917

SECTION B SUMMATIVE ASSESSMENT

REPORT 1: COMPUTER ARCHITECTURE ELEMENTS

TASK 1

1. Introduction

This report explains computer architecture elements using a practical example of a desktop computer (HP 280 G3, Intel Core i5, 8GB RAM, 500GB HDD) provided by the assessor. Computer architecture is the conceptual design and fundamental operational structure of a computer system.

2. Functions of elements which make up computer architecture

The main elements are:

1. Instruction Set Architecture (ISA) l: Functions as the interface between hardware and software. It tells the processor what operations it can perform.

2. Micro-architecture: functions as the implementation. It defines how the CPU is built to execute the ISA - pipelines, ALUs, caches.

3. system Architecture: Functions as the overall organization. It defines how CPU, memory, and I/O devices are interconnected via buses.

4. Memory System: functions to store data and instructions temporarily and permanently.

3.Functions with real-world application

a) ISA: function is to provide instructions. _Real world:_ On the assessed desktop, the x86-64 ISA allows it to run Windows 11 and Microsoft Office. On my smartphone, ARM ISA allows Android to run with low battery usage.

b) Micro-architecture: Function is processing efficiency. _Real world:_ The i5's micro-architecture has 4 cores and 6MB cache. When I rendered a past project video for college, the quad-core allowed multi-tasking without freezing.

c) System Architecture (Von Neumann): function is data movement. _Real world:_ In a supermarket POS system I observed during my practical, the same memory holds the scanning program and the price data.

D) memory Hierarchy: function is speed vs storage balance. _Real world:_ Registers (fastest) hold current calculation. In my current project, when I open a large Excel sheet, it moves from HDD to RAM for faster access.

4. Elements of computer architecture categories

Computer architecture has 3 main categories:

i) Instruction Set Architecture (ISA) Category: elements are Instruction Format, Data Types, Addressing Modes, Registers. Example: RISC vs CISC.

ii) Micro-architecture Category Elements are ALU (Arithmetic Logic Unit does maths), Control Unit (controls flow), Registers (small fast storage inside CPU), Cache Memory (L1, L2, L3), Pipeline.

iii) System Architecture Category: Elements are Motherboard, System Buses (Data, Address, Control), Main Memory (RAM), I/O Subsystem, Storage System.

5. Conclusion & Evidence

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REPORT 2: COMPUTER ORGANISATION

Task 2 - Name: Computer Organisation

1. Introduction

This is a follow-up of Task 1 using the same HP desktop computer. Computer organisation shows how the parts are physically connected and work together.

Identify/Name all components

Internal components identified: 1. Motherboard 2. CPU (Intel Core i5-8500) 3. RAM (DDR4 8GB) 4. Hard Disk Drive (HDD) / SSD 5. Power Supply Unit (PSU) 6. Graphics Processing Unit (GPU) 7. Network Interface Card 8. Cooling Fans.

External: Monitor, Keyboard, Mouse, Printer.

Purpose of each component

1. Motherboard: Backbone. Connects all parts and allows communication via chipset. Without it components cannot talk.

2. CPU: brain of computer. Fetches, decodes, executes instructions. Example: When I type my assignment, CPU processes each keystroke.

3. RAM: temporary workspace. Volatile. Purpose: Hold open programs. Example: Running Pastel accounting and Chrome at same time uses RAM.

4. HDD/SSD: Permanent storage for OS, files. Example: SSD stores my POE documents even when power is off.

5. PSU:Converts 220V AC from wall to 3.3V, 5V, 12V DC for components. Protects against power surges.

6. GPU: Processes graphics. Example: Displays my desktop on monitor and accelerates video playback.

7. I/O Devices: Input (keyboard/mouse) to give data, Output (monitor/printer) to get results.

HOW COMPUTER ELEMENTS WORK TOGETHER TO ALLOW DATA TO FLOW

Data flows through cooperation of Motherboard, CPU and Buses.

Practical example of booting the PC:

1. PSU powers motherboard.

2. CPU sends address via *Address Bus* on motherboard to BIOS chip.

3. BIOS data returns via *Data Bus*.
4. Control Bus sends READ signal.
5. CPU loads OS from SSD into RAM via Data Bus.
6. CPU executes OS instructions.

When I double-click Word, CPU requests Word's address, data moves from SSD -> RAM -> Cache -> CPU registers -> processed -> result sent via motherboard to monitor.

This shows: Motherboard = roads, Buses = vehicles carrying data, CPU = factory controlling traffic.

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REPORT 3: DESIGN CONSTRAINTS OF COMPUTER INSTRUCTION SETS

Task 3

1. Introduction

Instruction Set is limited by physical, cost and performance constraints.

Identify constraints

Six major constraints identified:

1. Instruction Length & Format Constraint: Fixed vs variable length.
2. Word Size Constraint: 32-bit vs 64-bit limits data processed at once.
3. Limited Number of Registers: Only 8-32 registers inside CPU.
4. Memory Capacity & Bandwidth Constraint: how much RAM CPU can address and how fast.
5. power Consumption & Heat Constraint: More complex instructions generate more heat and drain battery.
6. Cost and Compatibility Constraint: Design must be cheap to make and compatible with old software.
7. physical Size / Transistor Count Constraint

How constraints have been accommodated with practical examples

1. Accommodating Instruction Length: Designers use fixed-Length RISC for speed (ARM used in phones) and use hybrid approach for x86 - CPU translates complex x86 instructions into simple micro-ops inside. Example: Intel processors still accept old x86 instructions but internally use RISC-like execution to stay fast.

2. Accommodating Register Constraint: accommodated by compiler Optimization and Register Renaming. Compiler reuses registers smartly. Modern CPUs use register renaming to have 100+ physical registers even if ISA shows only 16. Example: In programming past project, Visual Studio compiler optimized C# code to use fewer registers.

3. Accommodating Memory Constraint: Accommodated by cache Memory, Virtual Memory and Pipelining. example: My 8GB RAM desktop runs a 20GB database project using virtual memory (using HDD as extra RAM) and L1/L2 cache reduces need to access slow RAM.

4. Accommodating Power Constraint: Accommodated by Low-Power ISA design (ARM), Clock Gating, and Sleep Modes. Example: Laptops use ARM or Intel U-series that lowers clock speed when not busy to save battery

5. Accommodating Compatibility Constraint: accommodated by backward Compatibility Extensions. Example: Intel added MMX, SSE, AVX instructions without removing old 8086 instructions, so Windows XP software still runs on Windows 11.